

GraySynth Algorithm^[1]

CNOT minimization in the presence of Z-basis rotations.

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Outline

1. CNOT-Phase Circuits
2. Building CNOT-Minimal Circuit
3. Example
4. Implementation

Definition: The *sum-over-paths* (f, T) of a circuit C over $CNOT$ and R_Z gates with associated unitary U_C is given by:

- a pseudo-Boolean function

$$f(\mathbf{x}) = \sum_{\mathbf{y} \in \mathbb{F}_2^n} \hat{f}(\mathbf{y}) \chi_{\mathbf{y}}(\mathbf{x}), \quad \chi_{\mathbf{y}}(\mathbf{x}) = x_1 y_1 \oplus x_2 y_2 \otimes \cdots \otimes x_n y_n$$

- a basis state transformation

$$T \in GL(n, \mathbb{F}_2^n)$$

such that

$$U_C = \sum_{\mathbf{x} \in \mathbb{F}_2^n} e^{2\pi i f(\mathbf{x})} |Tx\rangle \langle x|.$$

Sum-Over-Paths Form – Equivalence

Proposition: Given $f, f' : \mathbb{F}_2^n \rightarrow \mathbb{R}$ and $T, T' \in GL(n, \mathbb{F}_2)$, the unitary matrices

$$\sum_{\mathbf{y} \in \mathbb{F}_2^n} e^{2\pi i f(\mathbf{x})} |Tx\rangle \langle x|, \quad \sum_{\mathbf{y} \in \mathbb{F}_2^n} e^{2\pi i f'(\mathbf{x})} |T'x\rangle \langle x|$$

are equal if and only if $T' = T$ and $f' \sim f$.

Definition: The equivalence class of a phase function is defined as

$$[f] = \{f' : \mathbb{F}_2^n \rightarrow \mathbb{R} \mid f' = f + k, \text{ where } k : \mathbb{F}_2^n \rightarrow \mathbb{Z}\}.$$

Definition: A *parity network* for a set $S \subseteq \mathbb{F}_2^n$ is an n -qubit circuit C over CNOT gates where, for each $\mathbf{y} \in S$, the parity $\chi_{\mathbf{y}}(\mathbf{x})$ appears in the annotated circuit.

Definition: A parity network is *pointed at* $T \in GL(n, \mathbb{F}_2)$ if the overall transformation is T , i.e.

$$U_C = \sum_{\mathbf{x} \in \mathbb{F}_2^n} |T\mathbf{x}\rangle\langle\mathbf{x}|.$$

CNOT-minimal circuit

Minimal pointed parity network $\leftrightarrow$ CNOT-minimal circuit for a particular sum-over-paths form.

Equivalent circuits may have different sum-over-paths forms.

Optimize over all equivalent phase functions before optimizing the parity network.

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Given a set of binary strings S , synthesize a parity network up to some arbitrary overall linear transformation.

- Repeatedly choose an index j to expand.
- Recur on the co-factors S_0 and S_1 :

$$S_0 = \{\mathbf{y} \in S : \mathbf{y}[j] = 0\}, \quad S_1 = \{\mathbf{y} \in S : \mathbf{y}[j] = 1\}$$

- As a subset S' is expanded, apply CNOT gates where $\mathbf{y}[i] = 1$ iff $\mathbf{y}'[i] = 1, \forall \mathbf{y} \in S'$.

Lemma: Let C be a CNOT circuit and $S \subseteq \mathbb{F}_2^n$. For any positive integer i we let $C_{\leq i}$ denote the first i gates of C , c_i and t_i be the control and target of the i th CNOT gate in C . If we define

$$\begin{aligned} A_0 &= I & \mathbf{y}_0 &= \mathbf{y} \\ A_i &= E_{c_i, t_i} A_{i-1} & \mathbf{y}_i &= E_{t_i, c_i} \mathbf{y}_{i-1} \end{aligned}$$

for every $\mathbf{y} \in S$, then it follows that for any $\mathbf{x} \in \mathbb{F}_2^n$, $U_{C_{\leq i}}|\mathbf{x}\rangle = |A_i \mathbf{x}\rangle$ and

$$\chi_{\mathbf{y}_i}(A_i \mathbf{x}) = \chi_{\mathbf{y}}(\mathbf{x}).$$

The algorithm maintains the invariant that the remaining parities to be computed are expressed over the current bits.

Parity Network Construction - Pseudocode

Algorithm 1: Algorithm for synthesizing a parity network

```
1 function Gray-Synth( $S \subseteq \mathbb{F}_2^n$ ):
2   New empty circuit  $C$ ;
3   New empty stack  $Q$ ;
4    $Q.push(S, \{1, \dots, n\}, \epsilon)$ ;
5   while  $Q$  non-empty do
6      $(S, I, i) \leftarrow Q.pop$ ;
7     if  $S = \emptyset$  or  $I = \emptyset$  then return;
8     else if  $i \in \mathbb{N}$  then
9       while  $\exists j \neq i \in \{1, \dots, n\}$  s.t.  $y_j = 1$  for all  $y \in S$  do
10          $C \leftarrow C :: \text{CNOT}_{j,i}$ ;
11         for all  $(S', I', i') \in Q \cup (S, I, i)$  do
12           for all  $y \in S'$  do
13              $y \leftarrow E_{i,j} y$ ;
14        $j \leftarrow \arg \max_{j \in I} \max_{x \in \mathbb{F}_2} |\{y \in S \mid y_j = x\}|$ ;
15        $S_0 \leftarrow \{y \in S \mid y_j = 0\}$ ;
16        $S_1 \leftarrow \{y \in S \mid y_j = 1\}$ ;
17       if  $i \in \{\epsilon\}$  then
18          $Q.push(S_1, I \setminus \{j\}, j)$ ;
19       else
20          $Q.push(S_1, I \setminus \{j\}, i)$ ;
21        $Q.push(S_0, I \setminus \{j\}, i)$ ;
22   return  $C$ ;
```

- Define tuple (S, I, i) , where
 - S : subset of binary strings,
 - I : indexes left to be expanded,
 - i : index of CNOT's target qubit
- Data structure: LIFO stack Q

To synthesize a parity network pointed at T :

1. Synthesize a regular parity network, which produces linear transformation A ,
2. Invert the produced transformation and implement the required linear transformation, $TA^{-1} \rightarrow$ Patel-Markov-Hayes Algorithm [2]

To obtain the final circuit:

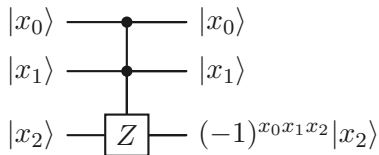
1. Build the circuit of the parity network,
2. Apply the Z-basis rotations where the associated parity is found in the circuit,
3. Apply the linear transformation TA^{-1} .

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Example – Setting

- CCZ: $|x\rangle \mapsto e^{2\pi i \frac{x_0 x_1 x_2}{2}} |x\rangle$



- Sum-over-path form (f, T) :

$$f(\mathbf{x}) = \frac{x_0 x_1 x_2}{2} = \frac{1}{8} \left[x_0 + x_1 + x_2 - (x_0 \oplus x_1) - (x_0 \oplus x_2) - (x_1 \oplus x_2) + (x_0 \oplus x_1 \oplus x_2) \right],$$

$$T = I$$

Example – Parity Network

Represent the set S as a matrix:

$$S = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ y_0 & y_1 & y_2 & y_3 & y_4 & y_5 & y_6 \end{pmatrix}$$

$$Q = [(S, \{0, 1, 2\}, \varepsilon)]$$

Select $j = 0$ and create co-factors S_0 and S_1 :

$$S_0 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ y_1 & y_2 & y_5 \end{pmatrix}, \quad S_1 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ y_0 & y_3 & y_4 & y_6 \end{pmatrix}$$

Example – Parity Network

$$Q = \left[\left(\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ y_0 & y_3 & y_4 & y_6 \end{pmatrix}, \{1, 2\}, 0 \right), \left(\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ y_1 & y_2 & y_5 \end{pmatrix}, \{1, 2\}, \varepsilon \right) \right]$$

Current subset S' :

$$S' = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ y_1 & y_2 & y_5 \end{pmatrix}$$

Select $j = 1$ and create co-factors S_0 and S_1 :

$$S'_0 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ y_2 \end{pmatrix}, \quad S'_1 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 1 \\ y_1 & y_5 \end{pmatrix}$$

Example – Parity Network

$$Q = \left[\left(\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ y_0 & y_3 & y_4 & y_6 \end{pmatrix}, \{1, 2\}, 0 \right), \left(\begin{pmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 1 \\ y_1 & y_5 \end{pmatrix}, \{2\}, 1 \right), \left(\begin{pmatrix} 0 \\ 0 \\ 1 \\ y_2 \end{pmatrix}, \{2\}, \varepsilon \right) \right]$$

Current subset S' :

$$S' = \begin{pmatrix} 0 \\ 0 \\ 1 \\ y_2 \end{pmatrix}$$

Select $j = 2$ and create co-factors S'_0 and S'_1 :

$$S'_0 = \emptyset, \quad S'_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ y_2 \end{pmatrix}$$

Example – Parity Network

$$Q = \left[\left(\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ y_0 & y_3 & y_4 & y_6 \end{pmatrix}, \{1, 2\}, 0 \right), \left(\begin{pmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 1 \\ y_1 & y_5 \end{pmatrix}, \{2\}, 1 \right), \left(\begin{pmatrix} 0 \\ 0 \\ 1 \\ y_2 \end{pmatrix}, \emptyset, 2 \right) \right]$$

Current subset S' :

$$S' = \begin{pmatrix} 0 \\ 0 \\ 1 \\ y_2 \end{pmatrix}$$

No need to add CNOT, since $y_2 = x_2$ is already on wire 3.

$$x_0 \text{ — } x_0$$

$$x_1 \text{ — } x_1$$

$$x_2 \text{ — } x_2$$

Example – Parity Network

$$Q = \left[\left(\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ y_0 & y_3 & y_4 & y_6 \end{pmatrix}, \{1, 2\}, 0 \right), \left(\begin{pmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 1 \\ y_1 & y_5 \end{pmatrix}, \{2\}, 1 \right) \right]$$

Current subset S' :

$$S' = \begin{pmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 1 \\ y_1 & y_5 \end{pmatrix}$$

Select $j = 2$ and create co-factors S_0 and S_1 :

$$S'_0 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ y_1 \end{pmatrix}, \quad S'_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ y_5 \end{pmatrix}$$

Example – Parity Network

$$Q = \left[\left(\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ y_0 & y_3 & y_4 & y_6 \end{pmatrix}, \{1, 2\}, 0 \right), \left(\begin{pmatrix} 0 \\ 1 \\ 1 \\ y_5 \end{pmatrix}, \emptyset, 1 \right), \left(\begin{pmatrix} 0 \\ 1 \\ 0 \\ y_1 \end{pmatrix}, \emptyset, 1 \right) \right]$$

Current subset S' :

$$S' = \begin{pmatrix} 0 \\ 1 \\ 0 \\ y_1 \end{pmatrix}$$

No need to add CNOT, since $y_1 = x_1$ is already on wire 2.

$$x_0 \text{ — } x_0$$

$$x_1 \text{ — } x_1$$

$$x_2 \text{ — } x_2$$

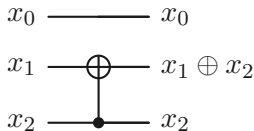
Example – Parity Network

$$Q = \left[\left(\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ y_0 & y_3 & y_4 & y_6 \end{pmatrix}, \{1, 2\}, 0 \right), \left(\begin{pmatrix} 0 \\ 1 \\ 1 \\ y_5 \end{pmatrix}, \emptyset, 1 \right) \right]$$

Current subset S' :

$$S' = \begin{pmatrix} 0 \\ 1 \\ 1 \\ y_5 \end{pmatrix}$$

Apply $\text{CNOT}_{2,1}$, then $y_5 = x_1 \oplus x_2$ is on wire 2.



Example – Parity Network

After $\text{CNOT}_{2,1}$, update according to: $\mathbf{y} \rightarrow E_{1,2}\mathbf{y}$, $\forall \mathbf{y} \in Q \cup S'$,

$$Q = \left[\left(\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ y_0 & y_3 & y_4 & y_6 \end{pmatrix}, \{1, 2\}, 0 \right) \right] \quad S' = \begin{pmatrix} 0 \\ 1 \\ 0 \\ y_5 \end{pmatrix}$$

Update matrix representing linear transformation applied by the circuit.

$$A_1 = E_{2,1}A_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Example – Parity Network

$$Q = \left[\left(\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ y_0 & y_3 & y_4 & y_6 \end{pmatrix}, \{1, 2\}, 0 \right) \right]$$

Current subset S' :

$$S' = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ y_0 & y_3 & y_4 & y_6 \end{pmatrix}$$

Select $j = 1$ and create co-factors S_0 and S_1 :

$$S'_0 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 1 \\ y_0 & y_4 \end{pmatrix}, \quad S'_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 0 \\ y_3 & y_6 \end{pmatrix}$$

Example – Parity Network

$$Q = \left[\left(\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 0 \\ y_3 & y_6 \end{pmatrix}, \{2\}, 0 \right), \left(\begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 1 \\ y_0 & y_4 \end{pmatrix}, \{2\}, 0 \right) \right]$$

Current subset S' :

$$S' = \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 1 \\ y_0 & y_4 \end{pmatrix}$$

Select $j = 2$ and create co-factors S_0 and S_1 :

$$S'_0 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ y_0 \end{pmatrix}, \quad S'_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ y_4 \end{pmatrix}$$

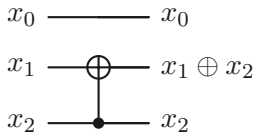
Example – Parity Network

$$Q = \left[\left(\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 0 \\ y_3 & y_6 \end{pmatrix}, \{2\}, 0 \right), \left(\begin{pmatrix} 1 \\ 0 \\ 1 \\ y_4 \end{pmatrix}, \emptyset, 0 \right), \left(\begin{pmatrix} 1 \\ 0 \\ 0 \\ y_0 \end{pmatrix}, \emptyset, 0 \right) \right]$$

Current subset S' :

$$S' = \begin{pmatrix} 1 \\ 0 \\ 0 \\ y_0 \end{pmatrix}$$

No need to add CNOT, since $y_0 = x_0$ is already on wire 1.



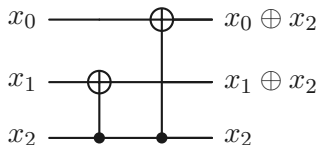
Example – Parity Network

$$Q = \left[\left(\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 0 \\ y_3 & y_6 \end{pmatrix}, \{2\}, 0 \right), \left(\begin{pmatrix} 1 \\ 0 \\ 1 \\ y_4 \end{pmatrix}, \emptyset, 0 \right) \right]$$

Current subset S' :

$$S' = \begin{pmatrix} 1 \\ 0 \\ 1 \\ y_4 \end{pmatrix}$$

Apply $\text{CNOT}_{2,0}$, then $y_4 = x_0 \oplus x_2$ is on wire 1.



Example – Parity Network

After $\text{CNOT}_{2,0}$, update according to: $\mathbf{y} \rightarrow E_{0,2}\mathbf{y}$, $\forall \mathbf{y} \in Q \cup S'$,

$$Q = \left[\left(\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}, \{2\}, 0 \right) \right] \quad S' = \begin{pmatrix} 1 \\ 0 \\ 0 \\ y_4 \end{pmatrix}$$

Update matrix representing linear transformation applied by the circuit.

$$A_2 = E_{2,0}A_1 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Example – Parity Network

$$Q = \left[\left(\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \\ y_3 & y_6 \end{pmatrix}, \{2\}, 0 \right) \right]$$

Current subset S' :

$$S' = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \\ y_3 & y_6 \end{pmatrix}$$

Select $j = 2$ and create co-factors S_0 and S_1 :

$$S'_0 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ y_3 \end{pmatrix}, \quad S'_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ y_6 \end{pmatrix}$$

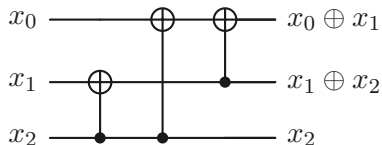
Example – Parity Network

$$Q = \left[\left(\begin{pmatrix} 1 \\ 1 \\ 1 \\ y_6 \end{pmatrix}, \emptyset, 0 \right), \left(\begin{pmatrix} 1 \\ 1 \\ 0 \\ y_3 \end{pmatrix}, \emptyset, 0 \right) \right]$$

Current subset S' :

$$S' = \begin{pmatrix} 1 \\ 1 \\ 0 \\ y_3 \end{pmatrix}$$

Apply $\text{CNOT}_{1,0}$, then $y_3 = x_0 \oplus x_2$ is on wire 1.



Example – Parity Network

After $\text{CNOT}_{1,0}$, update according to: $\mathbf{y} \rightarrow E_{0,1}\mathbf{y}$, $\forall \mathbf{y} \in Q \cup S'$,

$$Q = \left[\left(\begin{pmatrix} 1 \\ 0 \\ 1 \\ y_6 \end{pmatrix}, \emptyset, 0 \right) \right] \quad S' = \begin{pmatrix} 1 \\ 0 \\ 0 \\ y_3 \end{pmatrix}$$

Update matrix representing linear transformation applied by the circuit.

$$A_3 = E_{1,0}A_2 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

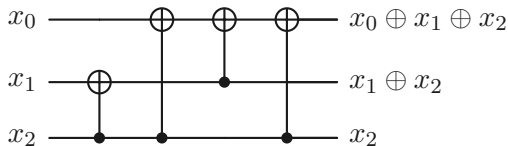
Example – Parity Network

$$Q = \left[\left(\begin{pmatrix} 1 \\ 0 \\ 1 \\ y_6 \end{pmatrix}, \emptyset, 0 \right) \right]$$

Current subset S' :

$$S' = \begin{pmatrix} 1 \\ 0 \\ 1 \\ y_6 \end{pmatrix}$$

Apply $\text{CNOT}_{2,0}$, then $y_6 = x_0 \oplus x_1 \oplus x_2$ is on wire 1.



Example – Parity Network

After $\text{CNOT}_{2,0}$, update according to: $\mathbf{y} \rightarrow E_{0,2}\mathbf{y}$, $\forall \mathbf{y} \in Q \cup S'$,

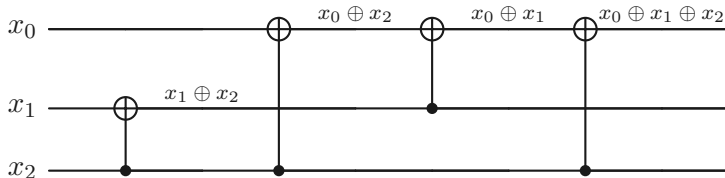
$$S' = \begin{pmatrix} 1 \\ 0 \\ 0 \\ y_6 \end{pmatrix}$$

Update matrix representing linear transformation applied by the circuit.

$$A_4 = E_{2,0}A_3 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

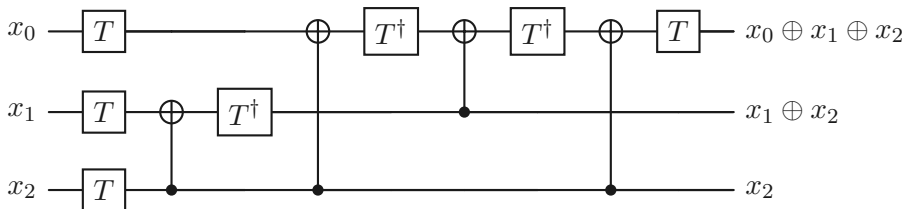
Example – Add Rotations

We apply the Z-basis rotations where the associated parity is found in the circuit.



Example – Add Rotations

Apply the Z-basis rotations where the associated parity is found in the circuit.



Example – Linear Transformation

Linear transformation produced by parity network is

$$A_4 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

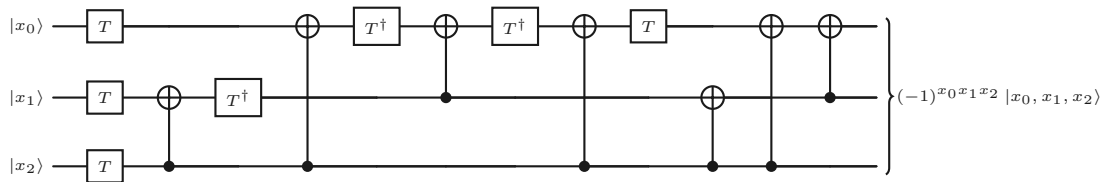
Required linear transformation is $T = I$

We apply A_4^{-1} inverting the decomposition of A_4 .

$$A_4^{-1} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \text{CNOT}_{1,0} \text{CNOT}_{2,0} \text{CNOT}_{2,1}$$

Example – Final Circuit

Combining the parity network with the corresponding Z-basis rotations and then uncomputing the parity network, the final circuit is



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Implementation

If you would like to check out the code, scan this:



Figure: <https://github.com/frafino/CNOT-Phase-Circuits-With-Minimal-CNOT-Count>

Thank you

Questions?

- [1] Matthew Amy, Parsiad Azimzadeh, and Michele Mosca. “On the controlled-NOT complexity of controlled-NOT–phase circuits”. In: *Quantum Science and Technology* 4.1 (2018), p. 015002. ISSN: 2058-9565. DOI: [10.1088/2058-9565/aad8ca](https://doi.org/10.1088/2058-9565/aad8ca). URL: <http://dx.doi.org/10.1088/2058-9565/aad8ca>.
- [2] K. N. Patel, I. L. Markov, and J. P. Hayes. *Efficient Synthesis of Linear Reversible Circuits*. 2003. arXiv: quant-ph/0302002 [quant-ph]. URL: <https://arxiv.org/abs/quant-ph/0302002>.